Chapter 12

Portfolio Theory and Asset Allocation

Chapter 2 introduced a two-asset portfolio to teach Python, and found, almost as an aside, that combining two negatively correlated assets produced a portfolio considerably less volatile than either asset alone. This chapter treats that observation as the central question of a whole field: given a set of assets, their expected returns, and how they move together, what is the best way to combine them?

Portfolio theory defines "best" quantitatively: the highest expected return for a given level of risk. This chapter develops the classical tools built on that definition (mean-variance optimization, the Capital Asset Pricing Model, and multi-factor models) along with the more robust methods portfolio managers use when the classical assumptions fail in real data.

Four worked examples run through the chapter. The two-asset AssetA/AssetB portfolio from Chapter 2 is extended into a full efficient frontier and a closed-form minimum-variance portfolio. The three-asset AssetA/AssetB/AssetC extension illustrates mean-variance optimization in matrix form and exposes a practical hazard the classical theory does not warn against: estimation error. A small market-model regression estimates from data the equity beta of 1.2 that Chapter 5's WACC calculation assumed. And a stylized equity/bond pair, with economically realistic volatilities and expected returns, carries the tangency, Black-Litterman, and robo-advisor examples, for which the four-day toy sample is too short and too oddly correlated.

A Brief History of Portfolio Theory

Knowing the sequence in which these ideas arrived clarifies which problem each was designed to solve. Table 12.1 summarizes the main milestones.

Year	Development
1952	Markowitz formalizes mean-variance optimization, giving diversification, already intuitive to practitioners, a precise mathematical foundation
1964	Sharpe (building on related work by Lintner and Treynor) introduces the Capital Asset Pricing Model, connecting an asset's expected return to a single market-wide risk factor
1990	Black and Litterman propose a method for blending an investor's own views with market-implied equilibrium returns, addressing a practical instability in naive mean-variance optimization
1992–1993	Fama and French introduce size and value factors alongside the market factor, launching multi-factor asset pricing models

Table 12.1: A brief history of portfolio theory

Each development responded to a limitation of what came before. Markowitz's framework requires estimating a full covariance matrix and is highly sensitive to estimation error, which motivated both the

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single-factor simplification of CAPM and, decades later, Black-Litterman's more stable blending approach. CAPM's single market factor, in turn, left systematic patterns in average returns unexplained, which Fama and French's additional factors were built to capture. This chapter follows roughly the same progression.

12.1 Diversification and the Efficient Frontier

Combining assets changes the risk of the whole in a way that does not follow from the parts, and that observation is the foundation of the chapter.

12.1.1 Diversification and the Power of Combining Assets

Chapter 2 computed the covariance matrix for AssetA and AssetB directly from their return histories,

$$\Sigma = \begin{pmatrix} 0.000414 & -0.000198 \\ -0.000198 & 0.000118 \end{pmatrix},$$

implying individual volatilities of $\sigma_A = 2.03\%$ and $\sigma_B = 1.09\%$ and a correlation of $\rho_{AB} = -0.896$. For a two-asset portfolio with weight w on Asset A and $1 - w$ on Asset B, portfolio variance is

$$\sigma_p^2(w) = w^2\sigma_A^2 + (1-w)^2\sigma_B^2 + 2w(1-w)\rho_{AB}\sigma_A\sigma_B. \tag{12.1}$$

Chapter 2 evaluated this formula only at the single equal-weighted point $w = 0.5$, finding $\sigma_p = 0.58\%$, considerably below the weighted average of the two individual volatilities.

Which value of w minimizes $\sigma_p^2(w)$? Differentiating with respect to w and setting the result to zero gives a closed-form solution,

$$w_A^{\text{MV}} = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}},$$

which, applied to the covariance matrix above, gives $w_A^{\text{MV}} = 34.05\%$ and $w_B^{\text{MV}} = 65.95\%$. This minimum-variance portfolio has volatility $\sigma_p^{\text{MV}} = 0.322\%$, nearly half the equal-weighted portfolio's 0.583%, with no new asset and no information about expected returns, only a better combination of the same two assets.

12.1.2 The Efficient Frontier and the Global Minimum-Variance Portfolio

Sweeping w across its full range traces out every risk-return combination achievable from AssetA and AssetB, illustrated schematically in Figure 12.1. Because $\rho_{AB} < 0$, the curve bows sharply to the left of a straight line connecting the two assets, the geometric signature of diversification: some combination is less risky than either asset alone, which is possible only because the assets do not move in lockstep.

In this sample, Asset B has *both* lower risk and higher return than Asset A ($\sigma_B = 1.09\%$ versus $\sigma_A = 2.03\%$, and $\bar{r}_B = 1.00\%$ versus $\bar{r}_A = 0.76\%$), so Asset A is not itself mean-variance efficient. The figure's upper arm, from the leftmost point of the curve to Asset B, is efficient in Markowitz's sense: every portfolio on it offers the highest expected return for its level of risk. The lower arm, from that point down to Asset A, is feasible but dominated, since a portfolio directly above it offers a strictly higher return at the same risk. The leftmost point is the *global* minimum-variance portfolio. The qualifier matters: every portfolio on the frontier is a minimum-variance portfolio for its own target expected return, whereas this one minimizes variance with no return constraint at all.

The efficient frontier is a menu rather than an answer. The framework narrows the feasible region down to the solid upper segment, and the investor picks a point on it according to how much variance they are willing to bear, a choice the mathematics leaves open. The global minimum-variance portfolio is the menu's left endpoint, the choice of an investor unwilling to accept any additional variance in exchange for return, not the portfolio the analysis points toward. It nonetheless earns a practical role: it is the only point on the frontier whose weights require no expected-return estimates, only the covariance matrix, which matters given how badly the estimation-error problem of Section 12.1.3 affects estimated means.

This is the quantitative content behind the advice to diversify: every portfolio off the efficient frontier is dominated by one on it.

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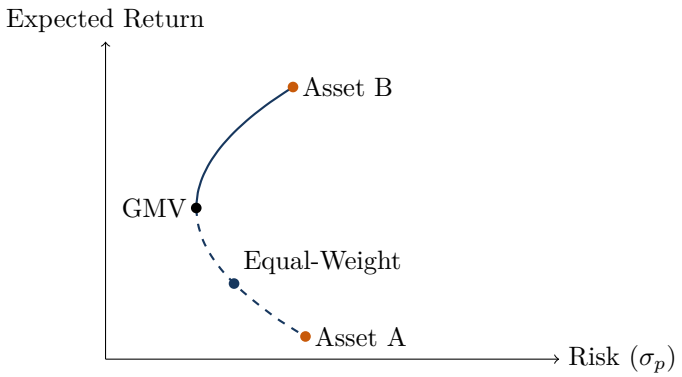


Figure 12.1: A schematic efficient frontier for the two-asset example (solid: efficient; dashed: feasible but dominated). GMV marks the global minimum-variance portfolio, the leftmost attainable point. An investor selects somewhere along the solid segment according to their risk tolerance; GMV is that menu's left endpoint, not its recommended item.

12.1.3 Mean-Variance Optimization in Matrix Form

The two-asset formulas above generalize to any number of assets using matrix notation. For n assets with covariance matrix Σ and a vector of ones $\mathbf{1}$, the (unconstrained) global minimum-variance portfolio weights are

$$w^{\text{MV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \tag{12.2}$$

the direct matrix generalization of the closed-form two-asset formula in Section 12.1.1. Applying this to the three-asset covariance matrix from Chapter 2's AssetA/AssetB/AssetC extension,

$$\Sigma_3 = \begin{pmatrix} 0.000414 & -0.000198 & -0.000383 \\ -0.000198 & 0.000118 & 0.000176 \\ -0.000383 & 0.000176 & 0.000358 \end{pmatrix},$$

gives minimum-variance weights of

$$w_A \approx 44.8\%, \qquad w_B \approx 14.2\%, \qquad w_C \approx 41.0\%,$$

with a portfolio standard deviation of only about 0.050%, roughly six times lower than the equal-weighted three-asset portfolio's 0.297% computed in Chapter 2.

This result deserves suspicion rather than celebration. Chapter 2's three-asset dataset contains only four daily observations, far too few to estimate a 3×3 covariance matrix (six distinct entries) reliably. The optimizer found weights that nearly cancel the sample's realized co-movements, but it is optimizing against estimation noise as much as against real risk. This is the overfitting Chapter 6 illustrated with a high-degree polynomial: with enough free parameters relative to the data, a model (here, a portfolio optimizer) can achieve an excellent in-sample result that will not repeat out of sample. It is Markowitz optimization's central weakness. In Michaud's phrase, the optimizer is an "estimation-error maximizer": it seeks out and overweights the asset pairs whose historical correlation was most favorable, which are disproportionately likely to be pairs whose favorable correlation was a statistical accident of the sample. Sections 12.2.5 and 12.2.6 introduce two standard responses.

12.1.4 Shrinkage Estimation of the Covariance Matrix

A more direct statistical response, distinct from risk parity and Black-Litterman, is to shrink the noisy sample covariance matrix toward a simpler, lower-variance target before handing it to the optimizer. A standard

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target is a diagonal matrix built from the average sample variance, and the shrunk covariance matrix is a weighted average of this target and the raw sample estimate,

$$\Sigma^{\text{shrunk}} = \delta F + (1-\delta)\Sigma, \qquad F = \bar{\sigma}^2 I,$$

where $\delta \in [0,1]$ is the shrinkage intensity (often chosen to minimize expected estimation error using a formula due to Ledoit and Wolf, though a fixed value illustrates the idea just as well) and $\bar{\sigma}^2$ is the average of the sample variances on the diagonal of Σ. Applying $\delta = 0.5$ to the three-asset covariance matrix from Section 12.1.3, whose average variance is $\bar{\sigma}^2 \approx 0.000297$, shrinks every off-diagonal covariance term roughly in half and pulls each diagonal variance partway toward the common average. Recomputing the minimum-variance weights using this shrunk covariance matrix gives

$$w_A^{\text{shrunk}} \approx 39.2\%, \qquad w_B^{\text{shrunk}} \approx 30.5\%, \qquad w_C^{\text{shrunk}} \approx 30.3\%,$$

a far more balanced allocation than the extreme weights the raw sample covariance matrix produced. The shrunk portfolio's sample variance is necessarily higher than the unshrunk optimizer's, which is the in-sample minimum by construction. The justification for shrinkage is that this modest in-sample cost buys a large improvement in out-of-sample reliability, a bias-variance trade-off analogous to the ridge regression penalty Chapter 6 applied to an overfit polynomial.

12.2 Equilibrium Models and Factor Approaches

If everyone optimizes, prices must satisfy conditions of their own; this part develops those conditions and the factor models that relax them.

12.2.1 The Capital Asset Pricing Model and Beta Estimation

Chapter 5 introduced the Capital Asset Pricing Model, $r_e = r_f + \beta(r_m - r_f)$, to estimate Meridian Fabrication's cost of equity, using an equity beta of 1.2 that was assumed rather than derived. Beta is a regression coefficient, estimated like the OLS regressions of Chapter 6 by regressing an asset's excess returns on the market's excess returns,

$$r_{i,t} - r_f = \alpha_i + \beta_i (r_{m,t} - r_f) + \varepsilon_t. \tag{12.3}$$

Table 12.2 shows six months of hypothetical excess returns for a stock and the overall market.

Month	1	2	3	4	5	6
Market excess return	−2.0%	−1.0%	0.0%	1.0%	2.0%	3.0%
Stock excess return	−2.5%	−1.3%	0.2%	1.1%	2.6%	3.3%

Table 12.2: Six months of excess returns for a stock and the market

Applying the OLS formulas,

$$\hat{\beta} = \frac{\sum_t (r_{m,t} - \bar{r}_m)(r_{i,t} - \bar{r}_i)}{\sum_t (r_{m,t} - \bar{r}_m)^2} \approx 1.19, \qquad \hat{\alpha} \approx -0.03\%, \qquad R^2 \approx 0.992.$$

The estimated beta of 1.19 closely matches the 1.2 that Chapter 5 assumed, a reassuring if constructed confirmation, and it shows where a CAPM beta comes from in practice: not from theory alone, but from an ordinary regression. The near-zero alpha is also economically meaningful. CAPM predicts that, after accounting for market exposure, no asset should offer a systematically positive or negative excess return, so a statistically significant nonzero alpha (checked with the t-statistic and p-value machinery of Chapter 6) is read either as evidence against CAPM or as a mispricing an active manager might exploit.

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Systematic versus Idiosyncratic Risk

The $R^2 \approx 0.992$ above also has a risk-decomposition interpretation. Total return variance splits into a systematic (market-driven) component and an idiosyncratic (asset-specific) component,

$$\mathrm{Var}(r_i) = \underbrace{\beta_i^2 \,\mathrm{Var}(r_m)}_{\text{systematic}} + \underbrace{\mathrm{Var}(\varepsilon_i)}_{\text{idiosyncratic}} \ ,$$

and R^2 is the fraction of total variance that is systematic. For the stock in Table 12.2, with returns expressed as decimals, $\mathrm{Var}(r_m) = 0.000350$ and systematic variance is $\hat{\beta}^2 \,\mathrm{Var}(r_m) \approx (1.1886)^2(0.000350) \approx 0.000494$ (carrying the unrounded slope $\hat{\beta} = 1.1886$; the rounded 1.19 would give 0.000496), against total variance of about 0.000498, leaving an idiosyncratic remainder of only about 0.0000038, consistent with the regression's near-perfect fit.

CAPM's central economic claim rests on this decomposition. Idiosyncratic risk can be diversified away by holding many assets whose asset-specific shocks are independent, so a well-diversified investor should not expect compensation for bearing it. Systematic risk, by contrast, affects every asset at once through the shared market factor and cannot be diversified away. That is why CAPM's expected-return formula compensates an asset only for its beta, not its total volatility. Two assets with identical total volatility but different betas should offer different expected returns, while two assets with identical betas but very different idiosyncratic volatility should offer the same expected return, a testable and occasionally violated prediction that motivated the multi-factor extensions later in this chapter.

The Security Market Line

The Security Market Line plots CAPM's expected return, $r_f + \beta(r_m - r_f)$, as a straight line against beta, shown in Figure 12.2. Unlike the Capital Market Line of Section 12.2.2, which plots return against total risk (σ_p) and is populated only by efficient portfolios, the Security Market Line plots return against systematic risk alone, and under CAPM every correctly priced asset or portfolio, efficient or not, lies on it.

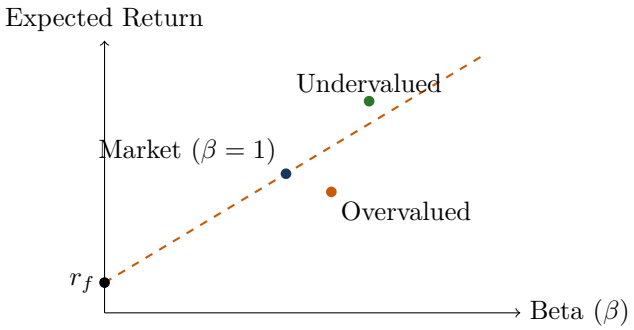


Figure 12.2: The Security Market Line plots CAPM's expected return against beta, with an undervalued asset (above the line) and an overvalued asset (below it) marked

An asset whose expected return plots above the Security Market Line is, under CAPM, undervalued: it offers more expected return than its systematic risk justifies, and the alpha from Section 12.2.1's regression is its vertical distance from the line. An asset below the line is overvalued. This is the graphical version of the active-management logic of the information ratio discussed later: an active manager searches for assets that plot persistently above the line. A manager who succeeds systematically is finding mispricing rather than taking on more systematic risk than the benchmark, a distinction the Security Market Line makes visible and a raw return comparison does not.

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12.2.2 The Capital Market Line and the Sharpe Ratio

CAPM's beta measures only one dimension of risk, co-movement with the market; the Sharpe ratio provides a complementary, portfolio-level summary of risk-adjusted performance,

$$\text{Sharpe Ratio} = \frac{\bar{r}_p - r_f}{\sigma_p}, \tag{12.4}$$

the excess return earned per unit of total risk taken. Comparing the equal-weighted and minimum-variance two-asset portfolios from Section 12.1.1, using their sample mean returns of 0.88% and 0.92% respectively and treating the daily risk-free rate as approximately zero for this short-horizon illustration,

$$\text{Sharpe}_{\text{equal}} = \frac{0.88\%}{0.583\%} \approx 1.51, \qquad \text{Sharpe}_{\text{MV}} = \frac{0.92\%}{0.322\%} \approx 2.85.$$

The minimum-variance portfolio has lower risk by construction, and here also a substantially higher Sharpe ratio, because it has a slightly higher mean return in this sample. That is not a general guarantee (the minimum-variance portfolio does not target return at all), but lower risk and competitive returns are not mutually exclusive.

A caution about units applies to every Sharpe ratio computed from the toy price series in this chapter. These are *daily* ratios, each estimated from four returns. Annualizing them by $\sqrt{252}$ would give about 24 and 45, numbers no real portfolio has ever sustained, not because these portfolios are extraordinary but because four observations cannot pin a Sharpe ratio down. For T independent returns, the standard error of an estimated Sharpe ratio is approximately $\sqrt{(1 + \text{SR}^2/2)/T}$ (Lo 2002), which here is 0.73 for the equal-weighted portfolio and 1.12 for the minimum-variance one, about as large as the estimates themselves. The calculation illustrates the formula; it is no evidence about either portfolio. Chapter 13's section on the cost of searching works out how much history a Sharpe ratio needs.

Mixing the risk-free asset with the tangency portfolio (the portfolio of risky assets with the highest Sharpe ratio) produces the Capital Market Line, illustrated in Figure 12.3: the set of risk-return combinations achievable from those two holdings.

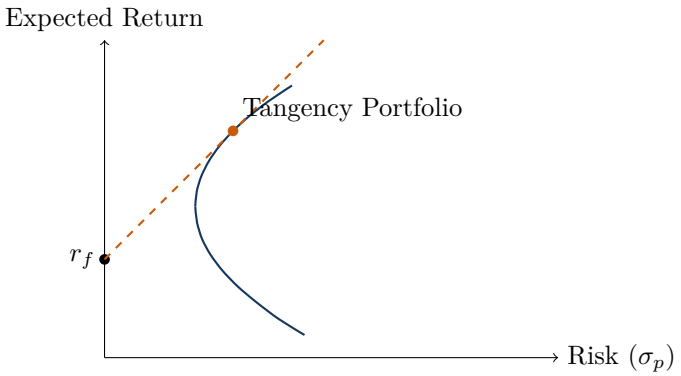


Figure 12.3: The Capital Market Line (dashed) is tangent to the efficient frontier of risky assets (solid) at the tangency portfolio

A central result of modern portfolio theory (the two-fund separation theorem) is that every investor, regardless of risk tolerance, should hold some combination of only the risk-free asset and this single tangency portfolio, differing in how much of each they hold, not in the composition of the risky portion. A more risk-averse investor holds more of the risk-free asset, while a less risk-averse investor may borrow at the risk-free rate to hold more than 100% of their wealth in the tangency portfolio, moving along the Capital Market Line without changing which risky assets they hold or in what proportion.

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12.2.3 Worked Example 12.1: Computing the Tangency Portfolio

The tangency portfolio of Figure 12.3 has a closed-form solution analogous to the minimum-variance formula, maximizing the Sharpe ratio rather than minimizing variance,

$$w^{\tan} = \frac{\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}, \tag{12.5}$$

Applying this to AssetA and AssetB, with mean daily returns $\mu_A = 0.76\%$, $\mu_B = 1.00\%$ and, as in Section 12.2.2, a risk-free rate of approximately zero, gives

$$w_A^{\tan} = 33.76\%, \qquad w_B^{\tan} = 66.24\%,$$

strikingly close to the minimum-variance weights of 34.05% and 65.95% computed in Section 12.1.1, with a Sharpe ratio of 2.849, barely above the minimum-variance portfolio's 2.848.

This near-coincidence is specific to this dataset. With $r_f \approx 0$ and both mean returns small relative to their risk, the excess-return vector $\boldsymbol{\mu} - r_f \mathbf{1}$ that drives the tangency formula is nearly proportional to a constant vector, so $w^{\tan}$ collapses close to the minimum-variance weights (driven purely by $\Sigma^{-1}\mathbf{1}$). Applying the same formula to the equity/bond assumptions of the robo-advisor discussion later in this chapter instead ($\sigma_e = 16\%$, $\sigma_b = 5\%$, $\rho_{eb} = -0.20$, $r_e = 8\%$, $r_b = 3\%$), with a more economically meaningful risk-free rate of $r_f = 2\%$, gives a tangency portfolio of

$$w_e^{\tan} = 32.0\% \text{ equities}, \qquad w_b^{\tan} = 68.0\% \text{ bonds}, \qquad \text{Sharpe} = 0.468,$$

meaningfully different from that section's minimum-variance weights of 13.1% equities and 86.9% bonds (Sharpe ratio 0.374 at the same risk-free rate). The two portfolios coincide only when expected returns do little to distinguish the assets relative to their risk. Whenever expected returns differ meaningfully, as for equities and bonds, the two can diverge substantially, and only the tangency portfolio is the one the two-fund separation theorem recommends.

12.2.4 Multi-Factor Models: Beyond a Single Market Beta

CAPM's single market factor leaves systematic patterns in average returns unexplained: smaller companies and companies with high book-to-market ratios have historically earned higher average returns than their betas predict. Fama and French's three-factor model extends CAPM with two additional factors, SMB (Small Minus Big, the return of small-capitalization stocks minus large-capitalization stocks) and HML (High Minus Low, the return of high book-to-market "value" stocks minus low book-to-market "growth" stocks), on top of the original market factor:

$$r_{i,t} - r_f = \alpha_i + \beta_{i,\text{MKT}}(r_{m,t} - r_f) + \beta_{i,\text{SMB}}\,\text{SMB}_t + \beta_{i,\text{HML}}\,\text{HML}_t + \varepsilon_t. \tag{12.6}$$

Estimating a two-factor version (market and SMB only) requires the matrix form of OLS from Chapter 6, $\hat{\beta} = (X^\top X)^{-1} X^\top y$. Fitting this to six months of hypothetical market, SMB, and asset return data gives

$$\hat{\alpha} \approx -0.0065\%, \qquad \hat{\beta}_{\text{MKT}} \approx 1.19, \qquad \hat{\beta}_{\text{SMB}} \approx 0.35, \qquad R^2 \approx 0.999;$$

the positive SMB loading indicates the asset behaves somewhat like a small-cap stock, earning a premium beyond what its market beta predicts.

The R^2 of 0.999 should not be read as an achievement. It is an artifact of how the six months were built: the asset's returns were generated from the two factors plus noise of only 0.067% a month, about one thirty-second of the asset's own 2.17% monthly volatility, and the regression has only $6 - 3 = 3$ residual degrees of freedom with which to discover otherwise. The same construction is behind the CAPM regression's R^2 of 0.992 earlier in this chapter. A real stock's idiosyncratic volatility is typically several percent a month, so a market-and-size regression on a single stock usually explains well under half of its variance, and a fit as tight as this one on real data would itself be a reason to look for an error. The estimated alpha is equally uninformative: its standard error, 0.039%, is six times its size ($t = -0.17$).

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Each additional factor is another explanatory variable in a multiple regression, so the overfitting cautions of Chapter 6 apply: adding enough factors always improves in-sample R^2. Serious factor research therefore relies on economically motivated factors validated out of sample, not an undisciplined search over every conceivable candidate, a practice derided in the literature as data mining for a "factor zoo."

Multi-factor models have moved into everyday investment products through "smart beta" or factor-based exchange-traded funds, which build portfolios with deliberately high exposure to a factor such as value, size, momentum, or low volatility, rather than weighting holdings by market capitalization. If a factor has historically earned a premium, an investor may prefer to capture it systematically and cheaply through a rules-based fund rather than paying an active manager for discretionary stock selection. The overfitting caution applies with particular force here. A factor premium discovered by statistical search can weaken or disappear once it becomes widely known and traded, as capital chasing the same exposure competes away the mispricing it compensated for. This is Chapter 3's market-efficiency argument: an exploitable pattern tends not to remain exploitable once it is traded at scale.

12.2.5 Inverse-Volatility Weighting and Risk Parity

Section 12.1.3 showed that full mean-variance optimization is dangerously sensitive to estimation error in the covariance matrix, particularly the correlation terms, which are the hardest to estimate. The simplest risk-based allocation, inverse-volatility weighting (sometimes loosely called naive risk parity), sidesteps much of this fragility by weighting inversely to volatility alone, ignoring correlation,

$$w_i^{\mathrm{IV}} = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}.$$

Applied to AssetA and AssetB,

$$w_A^{\mathrm{IV}} = 34.8\%, \qquad w_B^{\mathrm{IV}} = 65.2\%, \qquad \text{Sharpe} \approx 2.84,$$

remarkably close to the minimum-variance weights of 34.1% and 65.9% (Sharpe ratio 2.85) computed in Section 12.1.1, despite using no correlation estimate. That closeness is a feature of this dataset. For two assets, the minimum-variance weights coincide exactly with the inverse-volatility weights when the correlation is exactly -1, and AssetA and AssetB's sample correlation is -0.896, close to that limit. Holding the two volatilities fixed and raising the correlation shows how quickly the agreement breaks down: the minimum-variance weight on AssetA falls to 30.35% at $\rho = -0.5$, 22.18% at $\rho = 0$, and 2.41% at $\rho = 0.5$, while the inverse-volatility weight stays at 34.81% throughout. The case for inverse-volatility weighting is therefore robustness, not approximation: it requires estimating only n volatilities rather than $n(n-1)/2$ correlation pairs, each a source of the estimation error that made the three-asset optimization so unreliable. Risk parity proper, as practiced in industry, is usually the *equal risk contribution* (ERC) portfolio (Maillard et al. 2010), and unlike inverse-volatility weighting it does use correlations. Portfolio volatility splits exactly into one contribution per asset,

$$\sigma_p = \sum_i \mathrm{RC}_i, \qquad \mathrm{RC}_i = \frac{w_i(\Sigma w)_i}{\sigma_p}, \tag{12.7}$$

the same decomposition that underlies Chapter 8's Component VaR, and ERC chooses the weights that make every RC_i equal. Inverse-volatility weighting is the special case in which all pairwise correlations are equal. With only two assets the two methods give identical weights whatever the correlation, which is why AssetA and AssetB cannot tell them apart; the distinction first matters with three or more.

Worked Example 12.2: Equal Risk Contribution with Three Assets

Add a commodity sleeve to the equity/bond pair of Section 12.2.3: volatilities of 16%, 5%, and 20%, an equity-bond correlation of -0.20, equity-commodity 0.30, and bond-commodity 0.10. Inverse-volatility weighting gives 20%, 64%, and 16%, but the risk it produces is not balanced: by Equation (12.7), equities contribute 32.35% of portfolio risk, bonds 26.47%, and commodities 41.18%, because the commodity sleeve's positive correlation with equities adds risk that a volatility-only rule cannot see. (Equal weights are worse still: commodities then supply 58.77% of the risk and bonds 2.21%.)

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With three or more assets the ERC weights have no closed form and must be found numerically. A convenient route is that the ERC portfolio is the rescaled minimizer of the convex function $\frac{1}{2}\, y^\top \Sigma y - \frac{1}{n}\sum_i \ln y_i$ over positive y, whose first-order condition, $y_i(\Sigma y)_i = 1/n$ for every i, is the equal-contribution condition itself. Solving it gives

$$w_e^{\mathrm{ERC}} = 19.56\%, \qquad w_b^{\mathrm{ERC}} = 67.40\%, \qquad w_c^{\mathrm{ERC}} = 13.04\%,$$

for a portfolio volatility of 5.51%, against 5.90% for inverse volatility. Verifying the answer means applying the definition: at these weights, $w_i(\Sigma w)_i/\sigma_p^2$ returns 33.33% three times, and an independent solver that directly minimizes the squared differences between the contributions agrees to within 10^{-8}. ERC moved weight out of commodities and into bonds, the asset whose negative correlation with equities makes its marginal contribution to risk the smallest. For comparison, the minimum-variance portfolio of the same three assets holds 13.29%, 87.18%, and −0.46% for a volatility of 4.43%, with bonds supplying 87.18% of the risk: lower total risk, bought by concentrating it in one asset. ERC sits between the two, its usual position (Maillard et al. 2010): better balanced than inverse volatility, far less concentrated than minimum variance, and, like both, free of any expected-return estimate.

Accepting a method that is provably suboptimal under the classical assumptions in exchange for robustness when those assumptions fail is a recurring theme in practical portfolio management.

12.2.6 Black-Litterman: Blending Views with Market Equilibrium

Suppose you manage the 60/40 portfolio of Section 12.6.1 and you come to work with exactly one opinion: equities look cheap relative to bonds. This is the ordinary situation of a working investor, and the optimizer of Section 12.1.3 cannot accept it. That machinery demands a complete vector of expected returns, so before it will run you must also hand it an expected return for bonds, about which you have no opinion. Whatever number you invent, the optimizer treats it as a firmly held belief and acts on it as decisively as on the view you hold.

The second problem is that the formula has nowhere to record how sure you are. An opinion you would stake your reputation on and one you formed over coffee produce identical portfolios, though the difference between them ought to determine how far the portfolio moves from where it started.

Both problems land on the input that can least afford them. Chopra and Ziemba showed that, for a typical investor's risk tolerance, errors in estimated *means* are roughly an order of magnitude more costly, in utility lost, than errors of the same relative size in variances, which are in turn about twice as costly as errors in covariances. The expected-return vector is the most fragile input in the apparatus, and the naïve procedure requires you to fabricate most of it in order to express one real view. The same ordering is why the global minimum-variance portfolio of Section 12.1.2 is more robust than its extreme position on the frontier suggests: needing no expected returns, it is immune to the largest error source.

The Black-Litterman model answers both objections without touching the covariance matrix. It supplies a default answer to "what do I think about bonds?", so that holding no opinion becomes a position the mathematics can represent, and it lets every opinion carry an explicit statement of confidence.

The default comes from the market. Other investors' collective holdings are observable, and if they are broadly sensible they encode a set of expected returns: whatever numbers would have made that portfolio optimal to hold. Recovering them means running Section 12.1.3's machinery backward, a step called reverse optimization, which deduces the excess returns $\mathbf{\Pi}$ that make the observed market weights w_{mkt} optimal,

$$\mathbf{\Pi} = \delta\, \Sigma\, w_{\mathrm{mkt}}, \tag{12.8}$$

where δ is the market's coefficient of risk aversion. This answers the bond question with the number implied by what everyone else is already doing.

Your own opinions enter through three objects. A matrix P records which assets each view is about, so that "equities relative to bonds" can be expressed without saying anything about either alone. A vector Q gives the magnitudes claimed. And a matrix Ω states, on its diagonal, how uncertain each view is, where "I would stake my reputation on this" and "this is a hunch" become different inputs. The blend of prior and views is then

$$\mathbb{E}[R] = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\right]^{-1} \left[(\tau\Sigma)^{-1}\mathbf{\Pi} + P^\top \Omega^{-1} Q\right], \tag{12.9}$$

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the scalar τ setting how tightly the equilibrium prior is held. This is Chapter 1's Bayesian update in matrix form, with $\mathbf{\Pi}$ as the prior, Q as the evidence, and Ω^{-1} as the weight that evidence earns. A confident view (small Ω) pulls the answer toward Q; a tentative one leaves it near $\mathbf{\Pi}$.

Two consequences answer the opening problems. An investor with no views gets the market weights back unchanged, so "I have no opinion" produces a sensible portfolio rather than an arbitrary one. And an investor with a view, such as "equities will outperform bonds by 9% over the coming year," gets a portfolio tilted away from the market by an amount that depends on how firmly the view was asserted, rather than the lurch a historical-return optimizer would deliver. Black-Litterman thus combines a systematic baseline with analyst judgment that a purely statistical model like those in Chapter 6 cannot generate. The worked example that follows shows both consequences as numbers.

12.2.7 Worked Example 12.3: Blending a View into a 60/40 Portfolio

Take the equity and bond assumptions Section 12.2.3 already used ($\sigma_e = 16\%$, $\sigma_b = 5\%$, $\rho_{eb} = -0.20$, $r_f = 2\%$) and treat Section 12.6.1's classic 60/40 split as the market portfolio. That portfolio's volatility is 9.41% and, under the same 8% and 3% expected returns, its excess return is 4.00%, which pins down the risk-aversion coefficient as $\delta = 0.0400/0.0941^2 = 4.52$. Reverse optimization then implies equilibrium excess returns of

$$\Pi_e = 6.6546\%, \qquad \Pi_b = 0.0181\%.$$

Two features of this prior deserve attention. These are not the 8% and 3% assumed a moment ago, nor were they meant to be: they are the returns that would make a 60/40 investor's weights optimal, so the implied equity-minus-bond spread, $6.6546\% - 0.0181\% = 6.6365\%$, is wider than the 5.00% those assumptions embed. Bonds earn almost no equilibrium premium, not because they are unattractive but because their low volatility and negative correlation contribute almost nothing to the 60/40 portfolio's total risk, which is what a risk-based equilibrium ought to conclude.

Now suppose the investor holds the single view quoted above, that equities will beat bonds by 9%, against the 6.6365% equilibrium already implies. Setting $P = (1, -1)$, $Q = 9\%$, $\tau = 0.05$, and $\Omega = P(\tau\Sigma)P^\top$, which calibrates the view's uncertainty to the prior's own uncertainty about that same spread, the posterior expected excess returns become 7.68% for equities and -0.14% for bonds, and the resulting optimal weights are

$$w_e = 68.4\% \text{ equities}, \qquad w_b = 31.6\% \text{ bonds}.$$

The view moved the portfolio 8.4 percentage points toward equities rather than to a corner, and the size of that move tracks the confidence attached to it: halving Ω, which asserts the view more firmly, produces 71.1/28.9, while tripling it produces only 64.2/35.8. That bounded, monotone response is the point of the construction: the same 9% opinion yields a tilt the investor can dial rather than a violent swing, and setting the view aside returns exactly 60/40, the equilibrium the model began from.

12.3 Building and Running a Portfolio

A portfolio on paper becomes a portfolio in practice only after constraints, costs, currencies, and mandates are imposed on it.

12.3.1 Practical Allocation Decisions: Constraints, Rebalancing, and Transaction Costs

The portfolios developed so far ignore several frictions that matter enormously in practice. A long-only mandate prohibits negative weights (short positions), which the unconstrained formulas of Sections 12.1.1 and 12.1.3 do not rule out, and position limits or sector caps restrict concentration. Both typically require solving the optimization as a constrained quadratic program rather than with closed-form formulas.

A concrete example shows when the long-only constraint binds. Consider two assets with $\sigma_X = 8\%$, $\sigma_Y = 20\%$, and an unusually high correlation of $\rho_{XY} = 0.9$. Applying the closed-form minimum-variance formula from Section 12.1.1 gives

$$w_X^{\mathrm{MV}} \approx 145.5\%, \qquad w_Y^{\mathrm{MV}} \approx -45.5\%, \qquad \sigma_p \approx 5.26\%,$$

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a substantial short position in Asset Y financing a leveraged long position in Asset X, with volatility below either asset's own. A long-only mandate rules this out. Because portfolio variance decreases monotonically in w_X across the feasible range $w_X \in [0, 1]$ toward the (infeasible) unconstrained optimum, the best long-only portfolio is the corner solution $w_X = 100\%$, $w_Y = 0\%$, with Asset X's own 8% volatility. Here the constraint does not merely reduce the benefit of diversification; it eliminates it, since every feasible long-only combination is riskier than Asset X alone. The unconstrained formulas are a starting point for intuition, not a prescription: constraints can change the character of the optimal solution, not just its weights.

Even a well-constructed portfolio drifts from its target weights as assets earn different returns, so periodic rebalancing is necessary. Every rebalancing trade incurs the transaction costs of Chapter 3, at least half the bid-ask spread plus any market impact, so rebalancing that chases small deviations can destroy more value in trading costs than it preserves in risk control.

Consider a \$1,000,000 portfolio that has drifted 5 percentage points from its target weights, requiring a \$50,000 trade in each direction (selling the overweight asset, buying the underweight one) to restore them. At a one-way transaction cost of 15 basis points, each \$50,000 trade costs $\$50{,}000 \times 0.0015 = \$75$, for a total round-trip rebalancing cost of \$150, or 0.015% of the portfolio's value. That looks negligible once, but a manager who rebalances daily compounds it across hundreds of trading days per year. Practitioners therefore rebalance on a fixed calendar schedule, quarterly for example, or only when a weight drifts beyond a tolerance band, trading off precise tracking against cumulative cost drag.

Taxable investors face a further consideration: selling an appreciated position to rebalance can trigger capital gains tax, so a tax-aware strategy may tolerate more drift than risk minimization alone would recommend, in exchange for deferring the tax.

12.3.2 Solving the Constrained Problem Numerically

Once inequality constraints such as a long-only mandate or a position cap are added, the closed-form formulas of Sections 12.1.1 and 12.1.3 no longer apply, since setting a gradient to zero characterizes only an unconstrained (or purely equality-constrained) optimum. The general constrained minimum-variance problem is

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \, \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w}^\top \boldsymbol{\mu} = r_0 \text{ (optional)}, \quad w_i^{\text{low}} \le w_i \le w_i^{\text{high}} \ \ \forall i,$$

where the budget constraint ($\mathbf{w}^\top \mathbf{1} = 1$) is always present, the target-return constraint is included only when tracing a specific point on the efficient frontier, and the box constraint $w_i^{\text{low}} = 0$, $w_i^{\text{high}} = 1$ recovers a long-only mandate, a tighter w_i^{high} recovers a position or sector cap, and $w_i^{\text{low}} < 0$ permits limited short selling.

This quadratic objective with linear constraints is solved with a numerical constrained optimizer, most commonly sequential quadratic programming (SLSQP), available as `scipy.optimize.minimize(..., method="SLSQP")` in Python, which iterates toward a solution rather than computing one in a single matrix step.

Applied to the AssetX/AssetY example, minimizing variance subject to the budget constraint and long-only bounds $0 \le w_i \le 1$ converges in a handful of iterations to $w_X = 100\%$, $w_Y = 0\%$, the corner solution derived by hand, a useful check that the numerical method and the closed-form reasoning agree.

One practical wrinkle is easy to miss. A generic optimizer's default convergence tolerance is tuned for objective values of ordinary magnitude, and raw daily portfolio variances, on the order of 0.0001 or smaller, are small enough that a naive call can terminate after a single iteration at the starting point, silently reporting success despite having barely moved. Rescaling the objective (for instance, working in squared-percent units) or tightening the tolerance avoids this; the companion notebook shows the failure and the fix side by side.

This machinery also revisits Section 12.1.3's three-asset minimum-variance portfolio (weights of 44.8%, 14.2%, and 41.0% in AssetA, AssetB, and AssetC, with a volatility of about 0.050%) under a position cap, a common institutional constraint the closed-form formula cannot express. Capping every weight at 40% forces AssetA's allocation down from its unconstrained 44.8% to the 40% cap, with the freed-up weight redistributed to the other two assets:

$$w_A^{\text{cap}} = 40.0\%, \qquad w_B^{\text{cap}} = 28.4\%, \qquad w_C^{\text{cap}} = 31.6\%, \qquad \sigma_p^{\text{cap}} \approx 0.113\%,$$

more than double the uncapped portfolio's 0.050%, the price of the diversification the cap forces.

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Tracing a constrained efficient frontier under the same 40% cap adds the target-return constraint: solving for the minimum-variance portfolio achieving a 0.75% expected return, for instance, gives

$$w_A = 40.0\%, \qquad w_B = 35.1\%, \qquad w_C = 24.9\%, \qquad \sigma_p \approx 0.136\%,$$

one point on a constrained frontier that can be traced fully by sweeping the target return r_0 across a grid, as the companion notebook does.

12.3.3 Tracking Error and the Information Ratio

Many portfolio managers are evaluated relative to a benchmark, and the Sharpe ratio has a benchmark-relative analogue. The active return in any period is the portfolio's return minus the benchmark's return, $r_t^{\text{active}} = r_t^p - r_t^b$, and tracking error is the standard deviation of this active return series,

$$\text{Tracking Error} = \text{std}(r^{\text{active}}).$$

Using Chapter 2's equal-weighted AssetA/AssetB portfolio returns as the "portfolio" and AssetB's own returns as a hypothetical benchmark, the active returns across the four days are

$$r_1^{\text{active}} = 0.50\%, \qquad r_2^{\text{active}} = -1.24\%, \qquad r_3^{\text{active}} = 1.74\%, \qquad r_4^{\text{active}} = -1.48\%,$$

giving a tracking error of 1.52% and a mean active return of -0.12%. The information ratio, the active-return analogue of the Sharpe ratio, divides the two,

$$\text{Information Ratio} = \frac{\text{mean active return}}{\text{Tracking Error}} \approx \frac{-0.12\%}{1.52\%} \approx -0.08,$$

a small negative value indicating that, over this short illustrative sample, the portfolio slightly underperformed its benchmark per unit of active risk. A persistently negative information ratio suggests a manager is not adding value relative to holding the benchmark, net of fees. A positive and statistically significant one (checked with the t-statistic machinery of Chapter 6) is one standard piece of evidence that a manager's active decisions add value rather than reflect noise.

12.3.4 International Diversification and Currency Risk

Every example so far has assumed a single currency, but a portfolio that holds foreign assets adds currency risk: exchange-rate movements change the domestic-currency value of a foreign holding independently of how that asset performs in its local market. A US investor holding a European stock earns the stock's local-currency return combined with the euro's movement against the dollar, and the two can offset or reinforce each other.

International diversification is diversification applied across a wider opportunity set. An investor holding only domestic equities is implicitly betting that domestic conditions are the only risk worth diversifying, when returns across countries are far from perfectly correlated, since economies differ in industry mix, monetary policy, and business cycle. It is the mechanism of Section 12.1.1's two-asset example applied to national markets, and it is why most institutional portfolios hold meaningful international allocations despite the added currency risk.

A small numerical example shows how large the currency component can be. Suppose a US investor holds a European stock that returns 5% in euro terms over the period, while the euro simultaneously depreciates 3% against the US dollar. The unhedged US-dollar return combines both effects multiplicatively,

$$r_{\text{USD}}^{\text{unhedged}} = (1 + r_{\text{local}})(1 + r_{\text{FX}}) - 1 = (1.05)(0.97) - 1 \approx 1.85\%,$$

more than three percentage points below the stock's local-currency performance, purely because of currency movement. A fully hedged position, using the forward contracts of Chapter 3 to lock in today's exchange rate for the future conversion, would earn approximately the full 5% (less the hedge's cost), separating the stock-picking decision from the currency decision. Whether to hedge is itself a portfolio decision. Hedged exposure keeps the diversification benefit of foreign assets without the currency bet, at the hedge's cost; unhedged exposure retains the currency bet as an additional, sometimes uncompensated, source of risk and occasionally return. Institutional investors reach different conclusions about which suits their objectives.

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12.4 Machine Learning in Portfolio Construction

The classical methods in this chapter estimate their inputs (expected returns, variances, and covariances) with simple sample statistics, but each input is a prediction problem that machine learning can potentially improve. Regularized regression (ridge or LASSO) can estimate expected returns or factor exposures more stably than raw historical averages, shrinking noisy coefficients toward zero much as Section 12.1.4 shrank the covariance matrix.

12.4.1 Can Machine Learning Forecast Expected Returns?

Expected returns are the input whose estimation errors cost the most, so can the methods of Chapter 6 predict them better than a historical average? Few questions in empirical finance have been tested as thoroughly. Gu, Kelly, and Xiu ran the full comparison, putting ordinary least squares, penalized linear models, random forests, gradient-boosted trees, and neural networks on the same footing across roughly thirty thousand US stocks and ninety-four firm characteristics over six decades. Neural networks and tree ensembles won clearly, for the same reason they won in Section 12.4.2's much smaller example: they capture interactions and nonlinearities a linear specification cannot represent, and the linear benchmarks were close to useless out of sample.

The headline result is a monthly out-of-sample R^2 of roughly 0.40%, which a reader arriving from Chapter 6 will read as a catastrophic failure. It is not, and understanding why is the most transferable lesson in this section. Monthly equity volatility runs around 5% against an expected return near 0.7%, so noise dominates signal by construction, and an R^2 near half a percent is close to what is attainable. Campbell and Thompson made the argument precise: the right yardstick for a return forecast is not the fraction of variance it explains but the squared Sharpe ratio of the portfolio it funds, and by that standard a monthly R^2 of half a percent is economically substantial. Gu, Kelly, and Xiu's long-short portfolios formed on the neural-network forecasts duly earned Sharpe ratios far above what their linear competitors managed.

Three features keep the problem hard, none familiar from a standard machine learning course. The low signal-to-noise ratio is a property of the phenomenon, not the estimator. The target is adaptive and, in effect, adversarial: McLean and Pontiff found that documented anomaly returns shrink by roughly a quarter between a study's sample period and the period immediately after it, and by well over half once the result is published, so a pattern degrades because someone found it. And the data are scarcer than they look: sixty years of monthly observations amount to only a few hundred points per series, a thin diet for a method designed for millions of images.

Set beside Section 12.2.6's ordering, this yields the uncomfortable summary of the chapter's estimation problem: predictability runs opposite to importance. Variances and covariances, whose errors cost the least, are the most forecastable, which is why the volatility models of Chapter 7 work as well as they do. The level of expected returns, whose errors cost the most, is the least forecastable. In between sits cross-sectional prediction, deciding which assets will outperform which, which is materially more tractable than predicting the market's direction and is where nearly all successful applied work concentrates.

This also settles where an ML forecast should enter a portfolio process. Feeding it directly into the optimizer as $\boldsymbol{\mu}$ discards what makes it usable: the model reports not only a prediction but the accuracy of its past predictions. A forecast together with an out-of-sample error is the pair Black-Litterman asks for, a view Q and its uncertainty Ω, and routing it that way makes the tilt proportional to how much the model has earned. A predictor with a 0.40% R^2 receives a correspondingly large Ω and moves the portfolio only slightly, the correct response to a weak but real signal.

12.4.2 A Head-to-Head Test: Ridge Regression versus OLS on Collinear Factors

Section 12.2.4's two-factor example, with six months of data and two distinct factors, is too clean to show an estimation-error problem. Real candidate factors are often correlated (momentum and a "quality" factor both partly reflect recent price strength, say), and a short history makes that collinearity far more damaging. The companion notebook extends the factor regression to twelve months and five candidate factors (market, SMB, HML, momentum, and quality), with quality *constructed* as a near-linear combination of HML and momentum (0.61 and 0.89 correlation respectively) and with known true exposures $\beta = (1.10, 0.30, 0.20, -0.10, 0.15)$, so that each estimator's error can be measured against ground truth.

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Ordinary least squares, fit on all twelve months, badly overfits this short, collinear history:

$$\hat{\beta}_{\text{OLS}} = (1.078,\ -0.467,\ 0.979,\ 1.435,\ -2.322), \qquad \text{RMSE} = 1.390,$$

getting the market exposure roughly right but the SMB sign backward, wildly overstating HML and momentum, and estimating a quality exposure of -2.322 against a true value of $+0.15$. A ridge regression on the same data, with a small but nonzero penalty, recovers

$$\hat{\beta}_{\text{ridge}} = (0.957,\ 0.124,\ 0.027,\ -0.011,\ -0.075), \qquad \text{RMSE} = 0.167,$$

still imperfect, but with every coefficient pulled back to a plausible magnitude and no wild sign reversal, an 88% RMSE reduction against the five true exposures. This mirrors the covariance-shrinkage story of Section 12.1.4: the unregularized estimator is not merely less precise but misleading, fitting the collinear noise of one short sample, while a modest penalty trades a little bias for a large reduction in estimation error.

12.4.3 Beyond Return Forecasting: Hierarchical Risk Parity, End-to-End Learning, and Reinforcement Learning

Clustering methods, such as the K-means algorithm Chapter 6 applied to group firms by leverage and profit margin, can instead group assets by return co-movement before optimizing, an approach known as hierarchical risk parity. Rather than inverting a single, potentially ill-conditioned covariance matrix across all assets (the source of the fragility in Section 12.1.3), the portfolio is built recursively, allocating risk first across clusters of similar assets and then within each cluster. This tends to produce more stable, diversified allocations than full-matrix mean-variance optimization when the number of assets is large relative to the return history, a claim Section 12.4.4 tests out of sample.

More recent research trains tree ensembles and neural networks to output portfolio weights that maximize a risk-adjusted objective such as the Sharpe ratio directly, rather than first predicting returns and then optimizing. This end-to-end approach can avoid compounding prediction error with optimization error, but it inherits every overfitting and interpretability concern Chapter 6 raised, in a setting where an opaque model directly allocates capital, the model risk management concerns of Chapter 8. As elsewhere in this book, the message is not that these methods should be avoided but that they must be validated, monitored, and understood by the people responsible for the capital, not deployed because they are available.

A more experimental line of research frames allocation and rebalancing as a sequential decision problem and applies reinforcement learning: an algorithm learns an allocation policy through simulated trial and error against a reward such as risk-adjusted return net of transaction costs. This framing suits the rebalancing trade-offs discussed earlier, since an agent can learn to weigh close tracking of target weights against the cost of trading too often within a single objective. The caution applies with even greater force: an agent trained in a simulated market inherits every assumption of that simulation, and a policy that performs well there can fail unpredictably in real markets whose dynamics differ, an acute version of the non-stationarity concern raised throughout this book.

12.4.4 Worked Example 12.4: Hierarchical Risk Parity Out of Sample

Hierarchical risk parity (HRP) was introduced by López de Prado (2016), and since its justification is robustness, the test that matters is out of sample. The algorithm has three steps. First, it converts the estimated correlation matrix into a distance, $d_{ij} = \sqrt{(1-\rho_{ij})/2}$, which is 0 for perfectly correlated assets and 1 for perfectly anti-correlated ones, and clusters the assets hierarchically on that distance, merging the two closest assets or groups first, then the next closest, and so on (single linkage, in the language of Chapter 6's clustering material). Second, it reorders the assets in the order the resulting tree lists them, so that similar assets sit next to one another and the reordered covariance matrix is close to block-diagonal. Third, it allocates by *recursive bisection*: split the ordered list in half, give each half a variance V computed with inverse-variance weights inside it, divide the capital between the halves in the proportions $\alpha = 1 - V_{\text{left}}/(V_{\text{left}} + V_{\text{right}})$ and $1-\alpha$, and repeat inside each half until single assets remain. No matrix is ever inverted, which is the point.

The test universe is simulated, so that the true covariance matrix is known and every portfolio's out-of-sample risk can be measured exactly: twenty assets in four clusters of five, labeled A to D, with annual volatilities between 10% and 35%, correlations of 0.4 to 0.7 within a cluster, and 0.05 to 0.30 across clusters.

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On one 60-day estimation window, the clustering recovers the four groups exactly and lists cluster B, then A, then C, then D. The first bisection therefore separates A and B from C and D, with half-variances of 4.72×10^{-5} and 4.21×10^{-5} (daily units), which sends $\alpha = 47.1\%$ of the capital to the A and B half; the next two splits give 62.4% of that half to B and only 18.8% of the other half to C, the most volatile cluster. The final HRP weights run from 0.84% to 12.88%, all long. The sample minimum-variance portfolio built from the same 60 days puts 37.00% into a single asset and holds seven short positions totaling −62.55%. Measured against the true covariance matrix, its volatility is 9.34%, against 8.35% for HRP.

One window is an anecdote, so the companion notebook repeats the exercise a thousand times, each time estimating on 60 days and then holding the portfolio for the next 250. Table 12.3 reports the results.

Portfolio	Mean realized vol.	Std. dev.	95th pct. of true vol.
Equal weight	10.64%	0.46%	10.66%
Inverse variance	8.35%	0.39%	8.65%
Sample minimum variance	8.21%	0.64%	9.18%
Hierarchical risk parity	8.10%	0.42%	8.52%
Minimum variance, true covariance known		6.74%	

Table 12.3: Out-of-sample annualized volatility of portfolios estimated from 60 days of data and held for 250 days, over 1,000 simulated trials with twenty assets in four clusters. The standard deviation is that of realized volatility across trials; the last column is computed from the true covariance matrix.

HRP has the lowest average out-of-sample volatility, but only narrowly, 8.10% against 8.21% for the sample minimum-variance portfolio, and it beats that portfolio in only 54.8% of the individual trials. Its advantage is in the tail. The minimum-variance portfolio's risk varies across trials about half again as much as HRP's (0.64% against 0.42%), and in its worst twentieth of trials its true volatility exceeds 9.18%, against 8.52% for HRP: an optimizer that inverts a noisy matrix is sometimes excellent and sometimes badly wrong, while HRP is consistently decent. All four methods fall well short of the 6.74% available to an investor who knew the true covariance matrix, a direct measure of what estimating twenty volatilities and 190 correlations from 60 days costs.

The ranking is not fixed, either; it depends on how much data the optimizer is given. Averaged over 500 trials per window length, the sample minimum-variance portfolio's true volatility is 9.54% with 40 days of data, 8.23% with 60, 7.37% with 120, and 7.01% with 250, while HRP's barely moves, from 8.19% to 7.95%. HRP wins decisively when the number of assets is large relative to the history, ties at around 60 days, and loses once the history is long enough for the inverse to be trusted, because its bisection ignores the correlations across the halves it splits, information a well-estimated covariance matrix would use. That is the fair summary of HRP, and of every robust method in this chapter: it does not beat optimization, it beats optimization *on too little data*. In a real universe of hundreds of assets re-estimated every month, too little data is the ordinary situation.

12.5 Robo-Advisors: Automating Asset Allocation

A robo-advisor packages nearly every tool of this chapter (a risk-based target allocation, periodic rebalancing, tax awareness) into a single automated, low-cost retail product delivered through a web or mobile interface. Betterment and Wealthfront, both founded in 2008, pioneered the category; incumbents including Vanguard and Schwab entered a few years later, typically pairing the same automated core with an optional human advisor for a higher fee, a hybrid model this section returns to below.

The appeal is that the classical machinery, once encoded correctly, does not require a highly paid professional to execute, so automating it can extend a disciplined version of institutional-grade asset allocation to investors with far smaller balances than traditional advice has served.

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12.5.1 From a Risk Questionnaire to a Target Allocation

A new client typically completes a short questionnaire covering age, time horizon, income stability, and stated comfort with market declines, which the platform converts into a target stock/bond split. A common industry heuristic sets the equity share at 110 minus the investor's age, so a 35-year-old investor receives a suggested allocation of 75% equities and 25% bonds.

This chapter's machinery shows what the heuristic is and is not doing. Treating "equities" and "bonds" as two broad asset classes with illustrative volatilities $\sigma_e = 16\%$, $\sigma_b = 5\%$, and correlation $\rho_{eb} = -0.20$, Section 12.1.1's closed-form minimum-variance formula gives

$$\begin{aligned} \text{Minimum-variance:} &\quad 13.1\% \text{ equities},\ 86.9\% \text{ bonds},\ \sigma_p = 4.43\%, \\ \text{Questionnaire-based 75/25:} &\quad 75\% \text{ equities},\ 25\% \text{ bonds},\ \sigma_p = 11.81\%, \end{aligned}$$

the questionnaire split roughly two and a half times riskier than the minimum-variance point.

This is not a mistake in either number; it reflects two different objectives. Using illustrative expected returns of 8% for equities and 3% for bonds, and the same 2% risk-free rate as the tangency and Black-Litterman examples,

$$\begin{aligned} \text{Minimum-variance:} &\quad \mathbb{E}[r_p] = 3.65\%,\ \text{Sharpe} = 0.374, \\ \text{Heuristic 75/25:} &\quad \mathbb{E}[r_p] = 6.75\%,\ \text{Sharpe} = 0.402, \end{aligned}$$

so the heuristic improves on the minimum-variance point's risk-adjusted return. The minimum-variance portfolio ignores expected returns and holds far less equity than the 32.0%/68.0% tangency portfolio, whose Sharpe ratio of 0.468 is the highest these two assets can deliver. The 75/25 split overshoots the tangency weights in the other direction, with more than twice the tangency equity share, giving up some Sharpe ratio for substantially higher expected growth. A young investor with a long horizon and years of contributions ahead is pointed to a point beyond the tangency portfolio on the efficient frontier of Figure 12.1, a trade such an investor is usually willing to make.

This is a practical departure from the two-fund separation theorem, under which every investor holds the same tangency portfolio, using leverage if necessary to hold more than 100% of it. A robo-advisor does not offer a young client a leveraged position in a bond-heavy tangency portfolio; it shifts the mix of risky assets toward equity, because most retail investors neither can nor will borrow at the risk-free rate to lever up a conservative portfolio. As a client approaches a stated goal, typically retirement, the target allocation is walked back toward bonds along a predetermined glide path, the same lifecycle logic behind a conventional target-date fund.

12.5.2 Rebalancing and Tax-Loss Harvesting at Scale

The \$1,000,000 portfolio, 5-percentage-point drift, 15-basis-point cost example of Section 12.3.1 is the calculation a robo-advisor's rebalancing engine performs continuously across every client account, trading only once a client's drift tolerance band is breached rather than on a calendar schedule.

Tax-loss harvesting, a hallmark of the taxable (non-retirement) accounts these platforms offer, extends the automation to the client's tax bill. When a position falls in value, the platform sells it to realize the loss and immediately buys a similar, but not "substantially identical," security, avoiding the wash-sale rule that would disallow the loss, while keeping market exposure essentially unchanged.

Suppose a client holds a \$50,000 equity-fund position that has fallen 10%, a \$5,000 realized loss. Under U.S. tax rules, up to \$3,000 of a net capital loss can offset ordinary income in the year it is realized, and any remainder offsets capital gains. At a 32% marginal income-tax rate and a 15% long-term capital-gains rate, this single harvesting event saves $\$3{,}000 \times 32\% = \$960$ against ordinary income and $\$2{,}000 \times 15\% = \$300$ against capital gains, a first-year tax saving of \$1,260 on this one position. It is generated automatically and, across hundreds of thousands of accounts, far more consistently than a human advisor reviewing holdings once or twice a year could match. Some platforms extend this to direct indexing, holding an index's individual stocks rather than a single fund, which multiplies the tax lots available for harvesting at any time.

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12.5.3 Fee Structure and Long-Horizon Impact

A typical robo-advisor charges roughly 0.25% of assets annually, against roughly 1% for a traditional human advisor. Compounded over a long horizon, the gap is far larger than it appears. Starting from $100,000 and assuming a 7% gross annual return sustained for 30 years,

$$\begin{aligned} \text{0.25\% fee (robo-advisor):} &\quad \$709{,}637, \\ \text{1\% fee (traditional advisor):} &\quad \$574{,}349, \\ \text{0\% fee (hypothetical benchmark):} &\quad \$761{,}226. \end{aligned}$$

The lower-cost option preserves $135,288 more than the traditional-fee alternative over three decades, purely from a 75-basis-point difference in annual cost: fee drag through compounding, quantified with Chapter 5's time-value-of-money tools.

None of this makes a robo-advisor strictly superior to a human one. The risk questionnaire is a blunter instrument than a skilled advisor's judgment about a client's full financial picture; tax situations, estate planning, and major life events do not fit a set of multiple-choice questions. Self-reported risk tolerance is also unstable, shifting with recent market performance and question framing, so some platforms now refine a client's risk profile from ongoing behavior rather than treating the onboarding survey as permanent. This is where machine learning does distinct work in a robo-advisor. A classifier trained on a client's observed behavior, such as whether they reduced contributions or tried to sell equities in a past downturn, can produce a revealed-preference risk profile that predicts future behavior better than the questionnaire alone. An optimization layer decides, lot by lot and continuously, which tax lots to harvest and which substitute to buy, a large-scale constrained optimization no human reviewing accounts manually could replicate at that frequency. It is also why the industry has moved toward hybrid models, such as Vanguard's Personal Advisor Services and Betterment's premium tier, which layer a human advisor on the automated core for the situations the algorithm handles least well.

12.5.4 Case Vignette: Schwab Intelligent Portfolios and the Cost of Undisclosed Cash Drag

In June 2022, the U.S. Securities and Exchange Commission announced that Charles Schwab & Co. and an affiliated adviser would pay $187 million, $52 million in disgorgement and prejudgment interest and a $135 million civil penalty, to settle charges concerning its robo-advisor, Schwab Intelligent Portfolios. From March 2015 through November 2018, the SEC found, Schwab's algorithm allocated a materially large share of client portfolios to cash, averaging about 12.5% across accounts and ranging from roughly 6% for the most aggressive risk profiles to nearly 24.9% for the most conservative ones, without disclosing that Schwab's own internal analysis showed this allocation would very likely underperform a portfolio holding less cash under most market conditions. Schwab profited directly, earning an estimated $46 million from the spread between what it paid depositors on that cash and what it earned investing it, a consideration its marketing materials did not adequately disclose as shaping how the portfolios were built.

The episode illustrates a point Chapter 8's model-risk governance and Chapter 18's explainability material both made: an algorithmic allocation model is still a model, and a cash allocation is still an asset allocation decision, one this chapter's machinery would need to justify on risk-return grounds, not accept because the operator has a balance-sheet incentive to hold more of it. A robo-advisor's principal selling point is that automation removes the conflicts of interest and biases a commission-driven human advisor might introduce. That fails if the automation is built, silently, around the operator's own incentives. The governance discipline Chapters 8 and 15 developed for other financial models (independent validation, clear disclosure, and ongoing monitoring against the model's stated objective) therefore applies to a robo-advisor's allocation engine with no less force than to a bank's credit-scoring or trading model.

12.6 Institutional Allocation: Illiquidity and ESG Mandates

Large institutions face allocation questions a retail platform does not: horizons long enough to hold illiquid assets, and mandates that impose non-financial constraints on the portfolio.

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12.6.1 Case Vignette: From the 60/40 Portfolio to the Endowment Model

The history of asset allocation illustrates several of this chapter's themes. For decades, a simple 60% equities / 40% bonds allocation was the default institutional and individual portfolio, an informal application of the diversification logic of Section 12.1.1: equities and bonds have historically been imperfectly correlated (bonds have often, though not always, done relatively well in equity downturns), so a fixed mix captured much of diversification's benefit without any matrix algebra.

Beginning in the 1990s, several large university endowments, most prominently Yale's under David Swensen, moved toward what became known as the endowment model: substantial allocations to less liquid alternative assets (private equity, venture capital, hedge funds, real assets) and less in public equities and bonds. The logic applies this chapter's diversification and factor ideas: alternative assets have return drivers only partly explained by the market factor that drives public equities, so adding them can move a portfolio's position on the efficient frontier of Figure 12.1 favorably, not merely diversify within existing risk factors.

The benefit comes with a cost a mean-variance framework can understate: illiquidity. An endowment with a long horizon and predictable, modest spending can hold assets that cannot be sold for years, and be compensated with an illiquidity premium, a higher expected return for giving up the ability to trade. A pension fund or individual with nearer-term or less predictable liquidity needs faces a different trade-off and cannot copy the endowment model's allocations without its liquidity profile. That mismatch has caused real difficulty for institutions that adopted illiquid strategies without attention to their own funding needs.

12.6.2 ESG Constraints and the Cost of a Tilt

Many institutional investors now layer environmental, social, and governance (ESG) considerations onto asset allocation, either by excluding sectors or companies (negative screening) or by tilting weights toward higher-scoring companies within a sector (positive tilting), sometimes formalized as an additional factor alongside market, size, and value.

From a mean-variance perspective, any constraint or tilt not motivated by risk and expected return, including an ESG constraint, can only leave a portfolio on or below the unconstrained efficient frontier of Figure 12.1, never above it.

Whether ESG investing is nonetheless value-enhancing is disputed in the empirical literature and turns on questions this chapter's framework cannot settle. One is whether ESG characteristics proxy for otherwise-unpriced risks, in which case ESG exposure is a disguised risk factor capturable within the CAPM and multi-factor machinery. Another is whether investors will accept a lower expected return for values-alignment, a non-financial objective the mean-variance framework was never built to represent.

12.6.3 Worked Example 12.5: The Cost of an ESG Tilt

Section 12.3.2's numerical machinery makes the "on or below the efficient frontier" claim precise. Extend the three-asset AssetA/AssetB/AssetC covariance matrix with an illustrative ESG score for each asset on the 0-to-100 composite scale rating providers use: AssetA (a carbon-intensive legacy business) scores 40, AssetB (a clean-technology firm) scores 85, and AssetC (a diversified industrial) scores 60.

The unconstrained long-only minimum-variance portfolio, unchanged from Section 12.1.3, weights these 44.8%, 14.2%, and 41.0%, giving a portfolio ESG score of only $0.448(40) + 0.142(85) + 0.410(60) \approx 54.6$, dragged down by AssetA's large weight despite its low score.

Adding a minimum-portfolio-ESG-score constraint to the identical SLSQP optimizer, $\mathbf{w}^\top \mathbf{esg} \geq \text{ESG}_{\min}$, alongside the same budget and long-only constraints, reallocates weight away from AssetA and toward AssetB as the floor rises:

$$\begin{aligned}
\text{ESG} \geq 65: &\quad w = (36.0\%,\ 48.8\%,\ 15.2\%), &\quad \sigma_p \approx 0.216\%,\\
\text{ESG} \geq 70: &\quad w = (31.8\%,\ 65.4\%,\ 2.8\%), &\quad \sigma_p \approx 0.315\%,\\
\text{ESG} \geq 75: &\quad w = (22.2\%,\ 77.8\%,\ 0.0\%), &\quad \sigma_p \approx 0.484\%,
\end{aligned}$$

volatility rising from the unconstrained 0.050% to more than four times that at the mildest floor tested, and nearly ten times at the strictest. Every additional point of required ESG score is bought with additional variance, since the optimizer is no longer free to choose the covariance-minimizing weights.

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These ESG scores were assigned by hand to make the trade-off visible; a harder practical problem lies upstream, in choosing which score to use at all. Berg et al. (2022) compare ESG ratings for the same companies across six major providers and find an average pairwise correlation of only 38% to 71%, far lower than the near-perfect agreement credit rating agencies typically show for the same bond. The divergence traces mostly to disagreement about which metrics to measure and how to weight them, not to noisy measurement of an agreed concept.

A "high ESG" tilt built on one provider's scores can therefore differ meaningfully, in composition and in the volatility cost computed above, from one built on another's for the same universe: a model-risk concern inside the governance discipline of Chapters 8 and 18, applied to a data input rather than a fitted model.

12.7 Challenges in Portfolio Theory and Asset Allocation

Several challenges recur across nearly every practical application of this chapter's methods.

Estimation error in expected returns. Expected returns are far harder to estimate than volatilities or correlations, since return estimates are much noisier relative to their magnitude over any realistic sample. An optimizer fed noisy expected returns and a noisy covariance matrix compounds both errors, the problem Black-Litterman (Section 12.2.6) was designed to address.

Non-stationarity of correlations. As Chapter 8's discussion of the 2008 financial crisis noted, correlations can rise sharply in a crisis, exactly when diversification is needed most, so a portfolio built on calm-period estimates can protect far less in a crisis than the mathematics, calibrated on calm data, suggests.

Factor model instability. Factors that explain returns well in one period need not do so in the next, and multi-factor models, estimated on history, assume relationships that can and do fail as market structure and investor behavior evolve.

Crowding. When many investors independently adopt similar factor tilts or risk models, through smart-beta products, shared research, or convergent proprietary models, their positions can become highly correlated without anyone realizing it. A shock to one crowded strategy can then force mutually reinforcing selling across many ostensibly independent portfolios.

Liquidity and capacity constraints. Optimal weights may be unachievable for a large portfolio, since trading a large position in a less liquid asset moves its price, the market-impact concern of Chapter 3. A strategy that looks optimal on paper can underperform once realistic trading costs and capacity limits are included.

Behavioral and institutional deviations from the theory. Investors and institutions deviate from the optimal portfolio for reasons the classical theory does not capture: career risk (a manager who deviates from the benchmark and is wrong suffers far more than one who hugs it and is wrong), regulatory capital treatment of specific asset classes, and biases such as overconfidence or loss aversion.

Time-varying risk and return. Every formula in this chapter treats volatilities, correlations, and expected returns as constants estimated once from a sample. Chapter 7 showed that volatility clusters and evolves over time, and correlations and factor exposures likely shift too, so a framework that re-estimates its inputs infrequently can work from stale, sometimes badly wrong, assumptions.

Survivorship bias in factor research. Factor premiums are often estimated from databases that include only firms that survived to be recorded, quietly excluding bankruptcies and delistings. A strong historical premium may look much less attractive once those return-dragging failures are accounted for.

Multiple testing across the factor literature. Research has collectively tested an enormous number of candidate factors against the same limited history, so the multiple-testing concern of Chapters 6 and 7 applies to the published literature as a whole. Some factors will appear significant by chance alone, and clearing a significance threshold in one study does not make a factor exploitable.

Model risk in portfolio construction itself. Every optimizer, shrinkage estimator, and factor model in this chapter is a model requiring the governance discipline of Chapter 8 (independent validation, ongoing monitoring, documentation of known limitations), just like a bank's VaR or credit-scoring model. A process that treats its optimizer as a black box, rather than as one input checked against simpler benchmarks and economic intuition, is exposed to undetected model risk.

Liquidity mismatches. Any allocation to illiquid assets must be sized against the investor's own liquidity needs, not only the expected diversification benefit. An institution that mismatches the two can be

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forced to sell its most liquid holdings at the worst time because its illiquid ones cannot be sold, a dynamic related to the funding-liquidity risk of Chapter 8.

Benchmark and objective misspecification. The tracking error and information ratio of Section 12.3.3 are only as meaningful as the benchmark. A manager evaluated against a benchmark that does not reflect the mandate can appear to add or destroy value as an artifact of the comparison, a problem with no statistical fix, since the right benchmark is a judgment about what the portfolio is trying to achieve.

None of these challenges argues against the classical or robust methods; each is a reason to apply them with statistical rigor and institutional judgment, treating an optimizer's output as an informed starting point rather than a final answer.

12.8 Key Takeaways

Portfolio theory formalizes an intuition Chapter 2 first illustrated numerically: combining imperfectly correlated assets can reduce risk without sacrificing return, and mean-variance optimization finds the best combination, given accurate inputs. The efficient frontier traces every such combination; the Capital Market Line and two-fund separation theorem show that mixing the risk-free asset with a single tangency portfolio dominates every other combination; CAPM ties expected return to an estimable market beta (which Section 12.2.1 showed matches Chapter 5's WACC assumption) and splits total risk into systematic and idiosyncratic parts; and multi-factor models extend this logic to further systematic patterns.

The central practical warning is that full mean-variance optimization is acutely sensitive to estimation error, especially with limited data, which is why covariance shrinkage, risk parity, hierarchical clustering, and Black-Litterman exist as more robust, if theoretically suboptimal, alternatives. The sharpest form of that fragility is an inversion: the inputs rank in opposite order on how much their errors cost and how well they can be forecast. Covariances are the most predictable and least damaging to get wrong, expected returns the least predictable and by an order of magnitude the most damaging, which is why so much of the practical machinery avoids estimating expected returns at all.

Real portfolio management adds constraints that can qualitatively change optimal weights, rebalancing disciplined by transaction costs, tracking error and information ratios for benchmark-relative evaluation, and currency considerations, all of which the classical mathematics sets aside but a real allocation decision cannot ignore.

Robo-advisors automate this toolkit for retail investors: a risk questionnaire maps to a point on the efficient frontier, not necessarily the minimum-variance point; rebalancing and tax-loss harvesting run continuously at scale; and machine learning learns a client's revealed risk tolerance from behavior rather than a one-time questionnaire. The Schwab Intelligent Portfolios case showed that automating an allocation decision does not exempt it from the governance and disclosure discipline required of every other financial model.

12.9 Suggested Exercises

12.1 Using the two-asset formula in Section 12.1.1, verify by hand that $w_A^{\text{MV}} = 34.05\%$ minimizes portfolio variance, and compute the resulting portfolio's expected return.

12.2 Two assets have $\sigma_X = 15\%$, $\sigma_Y = 25\%$, and $\rho_{XY} = 0.20$. Compute the minimum-variance portfolio weights and the resulting portfolio volatility.

12.3 Explain, using Figure 12.1, why a rational investor would never hold a portfolio on the dashed (inefficient) segment of the frontier, even if they were extremely risk-averse.

12.4 Using the data in Table 12.2, verify by hand that the estimated beta is approximately 1.19, and compute the CAPM-implied cost of equity assuming $r_f = 3\%$ and a market risk premium of 7%.

12.5 A portfolio has an average daily return of 0.05% and a daily standard deviation of 1.1%. Assuming a daily risk-free rate of approximately zero, compute its Sharpe ratio, and compare it to the Sharpe ratios computed in Section 12.1.1.

12.6 Three assets have volatilities of 10%, 20%, and 30%. Compute their inverse-volatility weights.

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12.7 Explain, in your own words, why Michaud describes mean-variance optimization as an "estimation-error maximizer," referencing the three-asset example in Section 12.1.3.

12.8 Using Section 12.1.4's shrinkage formula and the three-asset covariance matrix of Section 12.1.3, recompute the minimum-variance weights at a shrinkage intensity of $\delta = 0.25$ rather than the $\delta = 0.5$ used in the text, and state whether they fall between the unshrunk weights and the $\delta = 0.5$ weights. Then explain, without computing anything further, what the weights must become as $\delta \to 1$, and why.

12.9 Describe a specific scenario in which an investor using Black-Litterman with no strong views would end up holding a different portfolio than one using naïve historical-mean mean-variance optimization, and explain why.

12.10 Using Section 12.2.7's equity/bond setup, recompute the Black-Litterman weights for an investor who believes equities will beat bonds by 7% rather than 9%, holding τ and Ω unchanged. Compare the resulting tilt away from 60/40 with the text's, and explain why a view only slightly above the equilibrium spread of 6.6365% moves the portfolio so little.

12.11 A portfolio manager rebalances a two-asset portfolio back to its 60/40 target weights every time the actual weight drifts by more than 5 percentage points. Explain, referencing Chapter 3's discussion of transaction costs, why this rule might outperform both a policy of rebalancing daily and a policy of never rebalancing at all.

12.12 Using Section 12.3.2's method, recompute the three-asset minimum-variance portfolio under a 50% (rather than 40%) position cap, and state whether the cap still binds on AssetA and, if so, by how much the resulting volatility differs from both the uncapped and the 40%-capped cases.

12.13 Explain, in your own words, why a numerical optimizer can terminate after a single iteration and report "success" on a badly scaled objective function, referencing Section 12.3.2's naive-versus-rescaled comparison, and describe one way to detect this failure mode besides comparing against a known closed-form answer.

12.14 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a mean-variance-optimized portfolio to perform poorly out of sample.

12.15 A portfolio earns returns of 1.5%, −0.5%, 2.0%, and 0.5% over four periods, while its benchmark earns 1.0%, 0.5%, 1.0%, and 0.5% over the same periods. Compute the tracking error and information ratio, and interpret the sign of the information ratio.

12.16 A US investor holds a Japanese bond that returns 2% in yen terms while the yen appreciates 4% against the US dollar over the same period. Compute the unhedged US-dollar return, and explain why it differs from the bond's local-currency return.

12.17 Explain, referencing the hierarchical risk parity idea in this chapter, why clustering assets before optimizing might produce more stable portfolio weights than applying full mean-variance optimization to all assets simultaneously.

12.18 Using the Security Market Line in Figure 12.2, explain what it means for an asset to plot above the line, and describe how this relates to the nonzero alpha estimated in Section 12.2.1.

12.19 Explain why a factor's historically estimated return premium might weaken after it becomes the basis of a widely traded smart-beta product, referencing the market-efficiency discussion in Chapter 3.

12.20 Using the robo-advisor discussion's method, compute the 110-minus-age heuristic equity weight for a 50-year-old investor, and, using the same illustrative $\sigma_e = 16\%$, $\sigma_b = 5\%$, $\rho_{eb} = -0.20$, $r_e = 8\%$, $r_b = 3\%$ assumptions with $r_f = 2\%$, compute that portfolio's volatility, expected return, and Sharpe ratio.

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12.21 A client holds a \$30,000 position that has fallen 8%. Using the robo-advisor discussion's tax-loss-harvesting method and a 24% marginal income-tax rate and 15% long-term capital-gains rate, compute the first-year tax savings from harvesting this loss.

12.22 Using the robo-advisor discussion's fee-drag example, recompute the 30-year future value of a \$100,000 initial investment at a 6% (rather than 7%) gross annual return for both the 0.25% and 1% fee cases, and state whether the dollar gap between them grows or shrinks relative to the text's example.

12.23 Explain, referencing Section 12.2.2's two-fund separation theorem, why a robo-advisor's practice of shifting a client's equity/bond mix based on risk tolerance is a departure from what that theorem, taken literally, would recommend.

12.24 Two assets have $\sigma_X = 12\%$, $\sigma_Y = 18\%$, $\rho_{XY} = 0.10$, $\mu_X = 9\%$, $\mu_Y = 6\%$, and $r_f = 1\%$. Using Section 12.2.3's formula, compute the tangency portfolio weights, and explain whether you would expect them to be closer to or further from the minimum-variance weights than the AssetA/AssetB example in the text, before actually computing the minimum-variance weights to check.

12.25 Explain, in your own words, why the tangency portfolio and the minimum-variance portfolio coincided so closely for AssetA and AssetB but not for the equity/bond example in Section 12.2.3, referencing the role r_f and the spread in expected returns each play in the tangency formula.

12.26 Using the three-asset equity/bond/commodity example of Section 12.2.5, verify with Equation (12.7) that the weights 19.56%, 67.40%, and 13.04% give each asset one third of the portfolio's risk. Then compute the risk shares of the equal-weighted portfolio (1/3 in each asset) and explain which asset dominates its risk, and why.

12.27 Section 12.4.4 found that hierarchical risk parity beats the sample minimum-variance portfolio with 40 days of data but loses to it with 250. Explain why the ranking reverses, and say which method you would expect to do better for a 500-stock universe re-estimated each month from one year (252 days) of daily returns, and why.

12.28 A strategy has a true daily Sharpe ratio of 0.10. Using the standard-error formula quoted in Section 12.2.2, how many daily observations are needed for the standard error of its estimated Sharpe ratio to fall to 0.05, half the quantity being estimated? Express the answer in years of 252 trading days, and compare it with the four observations behind this chapter's toy Sharpe ratios.

12.29 **Challenge (real data).** Download at least five years of monthly returns for five to ten real stocks (via `yfinance`) together with the Fama-French three-factor data for the same period (freely available from the Ken French Data Library). (a) Estimate each stock's CAPM beta by regressing its excess returns on the market factor, as in Section 12.2.1, and compare the range of estimated betas to the single illustrative beta of 1.19 used in this chapter. (b) Construct the minimum-variance and tangency portfolios from your assets' sample covariance matrix, as in Sections 12.1.1 and 12.2.3, and comment on whether the resulting weights look reasonable or exhibit the extreme, concentrated positions Michaud's "estimation-error maximizer" critique (Section 12.1.3) warns about. (c) Regress each stock's excess return on all three Fama-French factors (market, size, value) rather than the market alone, and discuss whether the market-only CAPM alpha from part (a) shrinks once size and value exposures are accounted for.

12.10 Further Reading

- Markowitz (1952), "Portfolio Selection." The original paper introducing mean-variance optimization, foundational to nearly every method in this chapter.
- Sharpe (1964), "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk." The original derivation of the Capital Asset Pricing Model used throughout Section 12.2.1.
- Fama and French (1993), "Common Risk Factors in the Returns on Stocks and Bonds." The original paper introducing the SMB and HML factors used in this chapter's multi-factor model example.

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- Black and Litterman (1992), "Global Portfolio Optimization." The original treatment of the Bayesian view-blending approach summarized in Section 12.2.6.
- Grinold and Kahn (2000), *Active Portfolio Management*. A practitioner-oriented treatment connecting mean-variance theory to real active-management decisions, including risk parity and other robust allocation methods introduced in this chapter.
- Bodie et al. (2024), *Investments*. A comprehensive textbook treatment of portfolio theory, CAPM, and factor models, extending this chapter's introductory coverage of each topic.
- Ang (2014), *Asset Management*. A modern, practitioner-oriented synthesis of factor models, smart beta, and the practical allocation considerations, including illiquidity and rebalancing, discussed later in this chapter.
- Michaud and Michaud (2008), *Efficient Asset Management*. The original source for the "estimation-error maximizer" critique of mean-variance optimization discussed in Section 12.1.3, along with resampling-based methods for producing more robust portfolios.
- Gu et al. (2020), "Empirical Asset Pricing via Machine Learning." The comprehensive machine learning bake-off behind Section 12.4.1, including the neural-network and tree results and the long-short portfolios built on them.
- Campbell and Thompson (2008), "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" The argument that a return forecast should be judged against the Sharpe ratio it produces rather than the variance it explains, which is what makes Section 12.4.1's tiny R^2 figures economically meaningful.
- McLean and Pontiff (2016), "Does Academic Research Destroy Stock Return Predictability?" The source of Section 12.4.1's post-publication decay estimates, and the clearest evidence that a return-forecasting target adapts to being modeled.
- Chopra and Ziemba (1993), "The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice." The source of Section 12.2.6's ordering of the three input-estimation errors by how much utility each destroys.
- Ledoit and Wolf (2004), "Honey, I Shrunk the Sample Covariance Matrix." The standard practitioner reference for Section 12.1.4's shrinkage estimator, including the formula for choosing the shrinkage intensity δ optimally rather than fixing it by hand as the text's example does.
- D'Acunto et al. (2019), "The Promises and Pitfalls of Robo-Advising." An empirical study of robo-advised accounts documenting the gap between clients' stated and revealed risk tolerance discussed in this chapter's treatment of robo-advisors.
- SEC (2022), *Schwab Subsidiaries Misled Robo-Adviser Clients about Absence of Hidden Fees*. The regulatory announcement underlying this chapter's Schwab Intelligent Portfolios case vignette.
- Berg et al. (2022), "Aggregate Confusion: The Divergence of ESG Ratings." The empirical study of cross-provider ESG rating disagreement underlying this chapter's ESG-tilt worked example.
- Maillard et al. (2010), "The Properties of Equally Weighted Risk Contribution Portfolios." The standard analysis of the equal-risk-contribution portfolio of Section 12.2.5, including why its volatility lies between those of the minimum-variance and equal-weighted portfolios.
- López de Prado (2016), "Building Diversified Portfolios that Outperform Out of Sample." The original description of hierarchical risk parity and of the Monte Carlo comparison that Section 12.4.4 reproduces in simplified form.
- Lo (2002), "The Statistics of Sharpe Ratios." The sampling theory of the Sharpe ratio behind Section 12.2.2's caution about the four-day toy ratios, including why ratios cannot in general be annualized by a square-root rule.

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